DISTRIBUTED ALGORITHMS • SPANNING TREES • DIJKSTRA PRIZE PAPER

Gallager-Humblet-Spira (GHS) Algorithm: Distributed MST

Synthesis based on Arpit Bhayani's Distributed Systems Masterclass • Gallager, Humblet, Spira (1983)

ALGORITHMIC PREMISE & DECENTRALIZED TREE OPTIMIZATION

Centralized MST algorithms (Prim, Kruskal) require global edge queues, making them impractical for distributed systems. The **Gallager-Humblet-Spira (GHS) algorithm** constructs the Minimum Spanning Tree across an arbitrary asynchronous network through autonomous, growing subtrees called **Fragments**. By iteratively identifying their **Minimum Weight Outgoing Edge (MWOE)** and merging across logarithmic levels, GHS achieves an optimal message complexity of $\mathcal{O}(|V| \log |V| + |E|)$, providing the theoretical foundation for Ethernet STP (IEEE 802.1D).

1. THE MINIMUM SPANNING TREE CHALLENGE

Given network $G = (V, E)$ with unique edge weights $w(e)$, an MST connects all $|V|$ nodes with $|V|-1$ edges minimizing $\sum w(e)$. In Layer 2 networks, an MST eliminates packet-destroying broadcast storms while maintaining full host reachability.

2. FORMAL SYSTEM MODEL & EDGE CLASSIFICATION

Network Model

- **Connected Undirected Graph:** $G = (V, E)$ with $|V| = n, |E| = m$.
- **Distinct Edge Weights:** Ensures a unique MST.
- **Asynchronous FIFO:** Communication is reliable but message delays are non-deterministic.

Three Edge States (Node u)

- `BASIC` : Unexplored, unclassified link.
- `BRANCH` : Confirmed edge in the final MST.
- `REJECTED` : Internal non-tree edge connecting two nodes in the same fragment. Discarded.

3. THE CUT PROPERTY & MWOE INVARIANT

The Cut Property: For any partition cut $(S, V \setminus S)$ of a graph, the strictly minimum weight edge crossing the cut must belong to the Minimum Spanning Tree.

Every Fragment in GHS forms an isolated cut. Consequently, when a fragment finds its Minimum Weight Outgoing Edge (MWOE) connecting to an outside node, that edge is guaranteed to be part of the global MST.

4. MWOE DISCOVERY & EDGE TESTING

To locate the fragment's MWOE without global coordination:

1. The core edge broadcasts a `FIND(Level, FragmentID)` wave downward through all `BRANCH` edges.
2. Each node u tests its lowest-weight `BASIC` edge via a `TEST(Level, FragmentID)` query.
3. The neighbor v responds:
   - If $ ext{FragmentID}_v == ext{FragmentID}_u$: Internal edge! Node u marks it `REJECTED` and tests the next edge.
   - If $ ext{FragmentID}_v eq ext{FragmentID}_u$ and $ ext{Level}_v \ge ext{Level}_u$: Confirmed outgoing edge! Replies `ACCEPT`.
   - If $ ext{Level}_v < ext{Level}_u$: Defers response until v's level catches up.
4. Each node reports its lowest confirmed outgoing edge upstream via `REPORT(weight)` convergecast to the core.

5. FRAGMENT MERGING & LEVEL ELEVATION RULES

When Fragment F_1 at Level L_1 contacts Fragment F_2 at Level L_2 across MWOE e:

Scenario	Protocol Action	New Level	Structural Impact
Absorption ($L_1 < L_2$)	F_1 sends `CONNECT`. F_2 absorbs F_1.	Remains L_2	Small fragment integrates into larger fragment without level elevation.
Combination ($L_1 = L_2$)	Both fragments mutually select edge e as MWOE!	**Increments to $L + 1$!**	Fragments unite across e; edge e becomes the new Core Edge.
Postpone ($L_1 > L_2$)	F_2 postpones processing until its level catches up.	No change	Prevents low-level fragments from stalling high-level progress.

6. FORMAL COMPLEXITY PROOF

Level Bound: $\mathcal{O}(\log |V|)$

A fragment elevates to level $L+1$ only when two fragments of level L combine. A level L fragment has at least 2^L nodes. Thus: $2^L \le |V| \implies L_{\max} \le \lfloor \log_2 |V| floor$.

Message Bound: $\mathcal{O}(|V| \log |V| + |E|)$

Control messages (`FIND`, `REPORT`) generate $O(|V|)$ per level $\implies \mathcal{O}(|V| \log |V|)$. Each edge is tested and rejected at most once $\implies \mathcal{O}(|E|)$. Total is optimal.

7. REAL-WORLD APPLICATIONS: ETHERNET SPANNING TREE (STP)

GHS directly inspired the IEEE 802.1D Spanning Tree Protocol (STP) deployed across global switching backbones:

Broadcast Storm Elimination

Without an MST, redundant Ethernet links circulate broadcast packets endlessly, crashing switch CPUs. STP disables redundant ports (marking them `REJECTED`/Blocking) to maintain an acyclic tree.

Dynamic Link Failover

If a primary fiber link is severed, the network fragments. Switches detect BPDU timeout, unblock standby ports, and recalculate the MST in under 2 seconds (RSTP: < 50 ms).

8. COMPARATIVE ANALYSIS: CENTRALIZED VS. DISTRIBUTED MST

Algorithm	Execution Model	Requires Global Visibility?	Time Complexity	Message / Work Complexity
Kruskal's Algorithm	Centralized	Yes (Global edge sort)	$\mathcal{O}(\|E\| \log \|V\|)$	In-memory disjoint-set union
Prim's Algorithm	Centralized	Yes (Global priority queue)	$\mathcal{O}(\|E\| + \|V\| \log \|V\|)$	Single priority queue
GHS Algorithm	Distributed (Async)	**No (Local edge testing)**	$\mathcal{O}(\|V\| \log \|V\|)$	**$\mathcal{O}(\|V\| \log \|V\| + \|E\|)$ messages**

9. SYSTEMS ENGINEERING TAKEAWAYS

1. **Parallel Component Merging:** GHS proves that global optimization does not require global synchronization. Independent clusters can make locally optimal decisions that guarantee global correctness.
2. **Level Invariant for Asynchrony:** In asynchronous networks where link delays vary wildly, numerical levels protect fast-moving fragments from being locked or corrupted by slow lagging nodes.
3. **Edge State Caching:** Permanently marking internal edges as `REJECTED` ensures that non-tree links are never re-evaluated, bounding the topological search overhead to $\mathcal{O}(|E|)$.